



Cloud Services

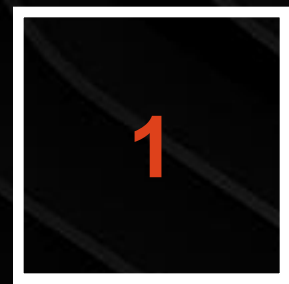
REGIONAL SCHOOL OF DEVOPS PROGRAM



Agenda

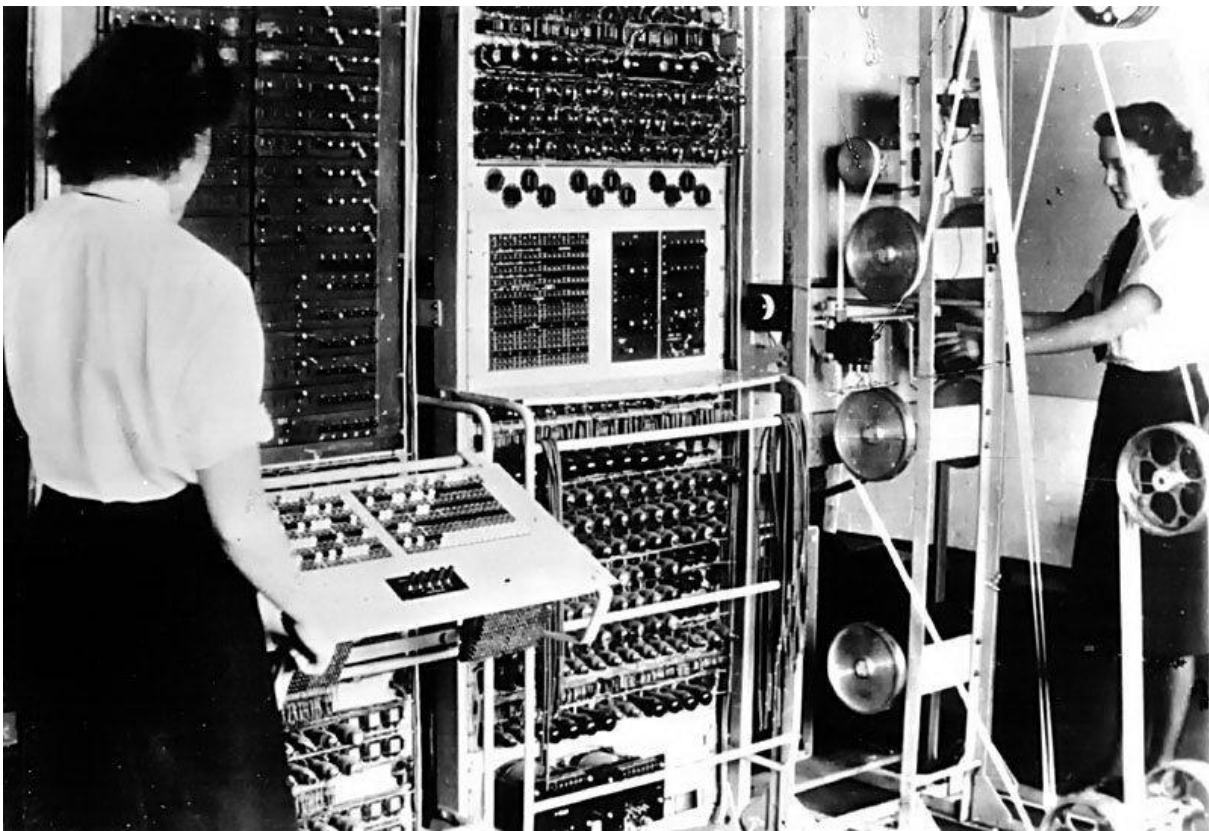
1. DAWN OF COMPUTING
2. VIRTUALIZATION
3. CLOUD PROPERTIES
4. • PRIVATE, PUBLIC AND HYBRID CLOUD
5. • IAAS, PAAS, SAAS
6. • ELASTICITY
7. • SCALABILITY
8. • HIGH AVAILABILITY
9. • FAULT TOLERANCE
10. • DISASTER RECOVERY
11. • REGIONS AND AVAILABILITY ZONES
12. RESOURCES



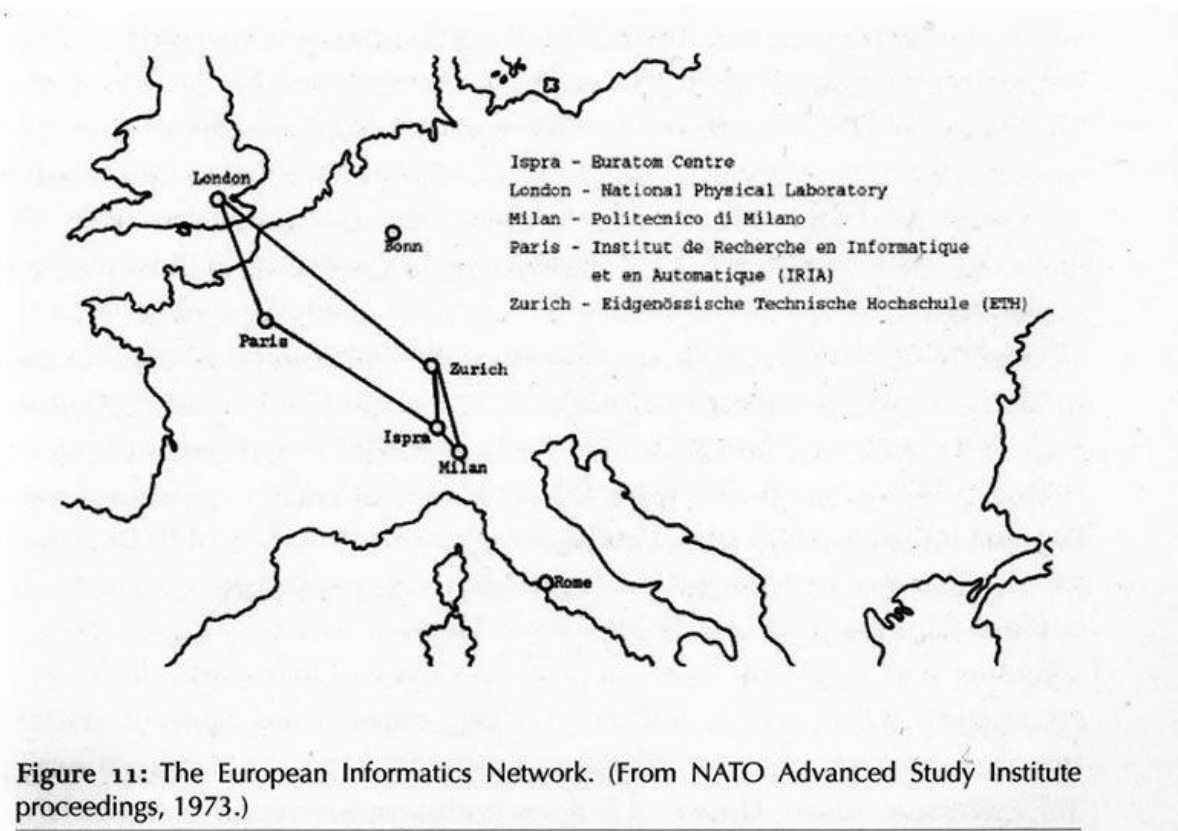


ONLY TODAY WAS NOT YESTERDAY

Dawn of computing



1943 COLOSSUS
BRITISH ENGINEER TOMMY FLOWERS

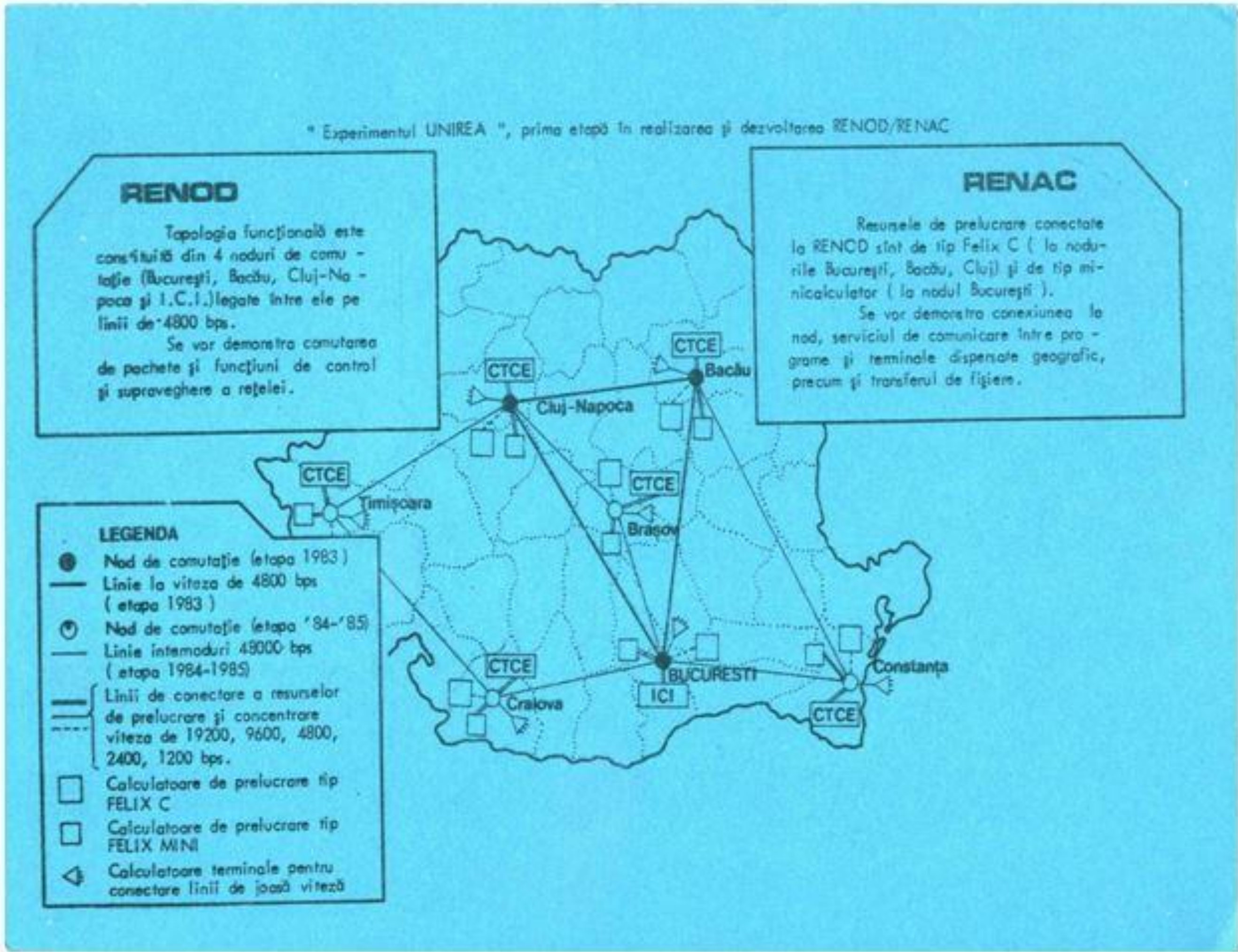


1973 SEAGATE ST506
FRANCE'S CYCLADES AND
BRITAIN'S NPL NETWORK



1980 SEAGATE ST506
THE DISK HELD 5 MEGABYTES OF DATA

1984: EXPERIMENTUL UNIREA

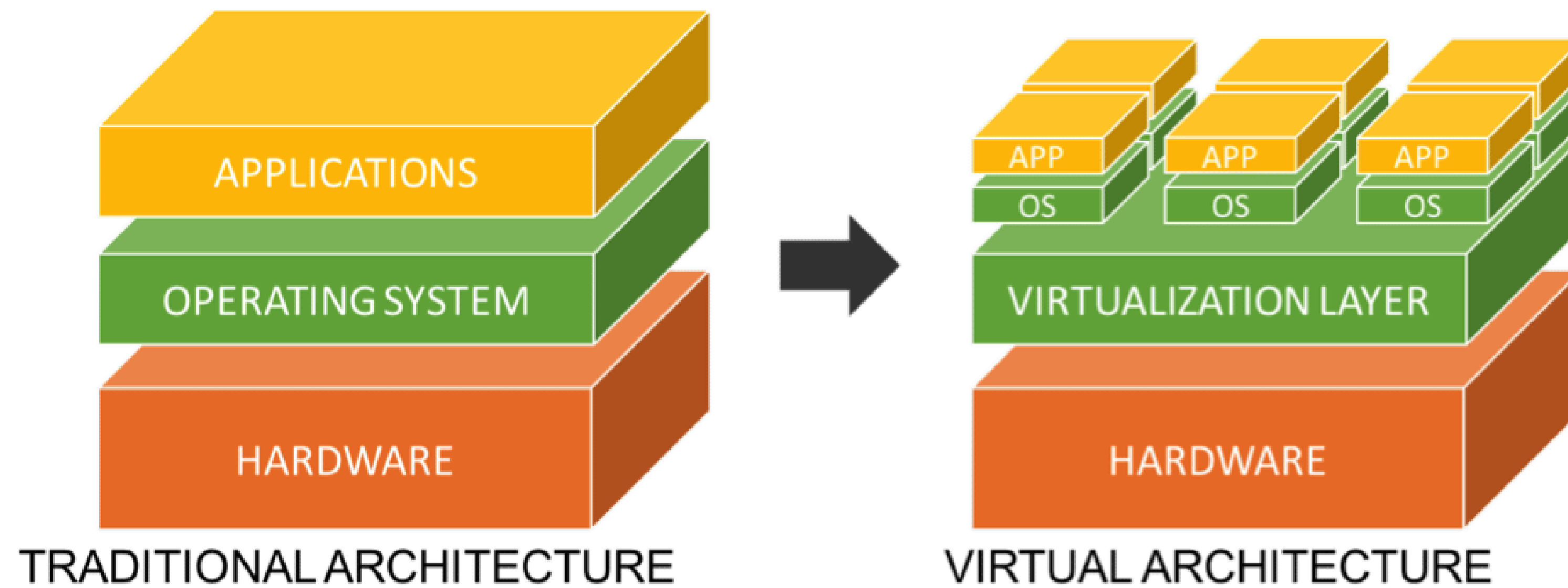




IT WOULD BE EASIER TO EXPLAIN SCHRÖDINGER'S CAT PHILOSOPHY THAN VIRTUALIZATION
...THUS, PLEASE SHED A TEAR

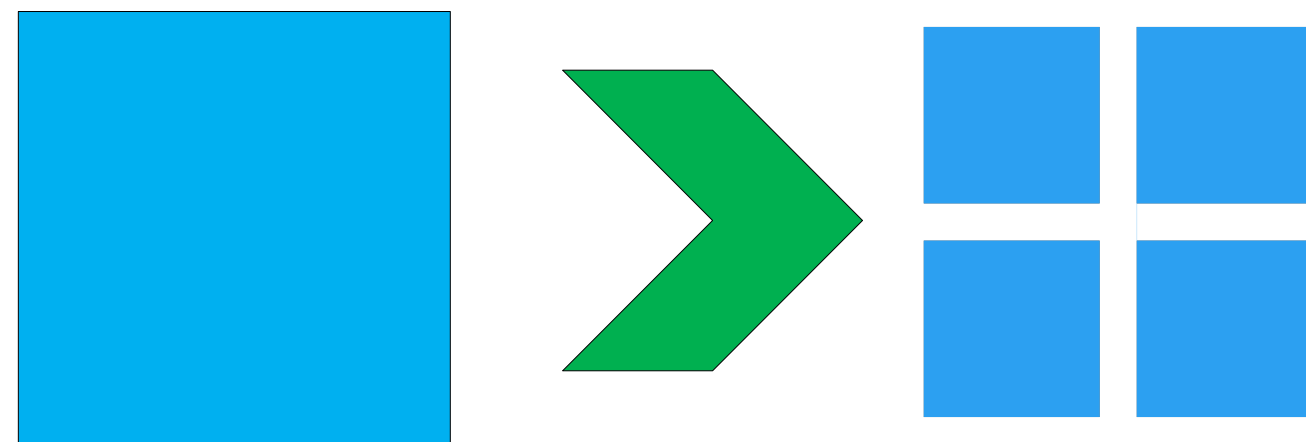
Virtualization

System Architecture

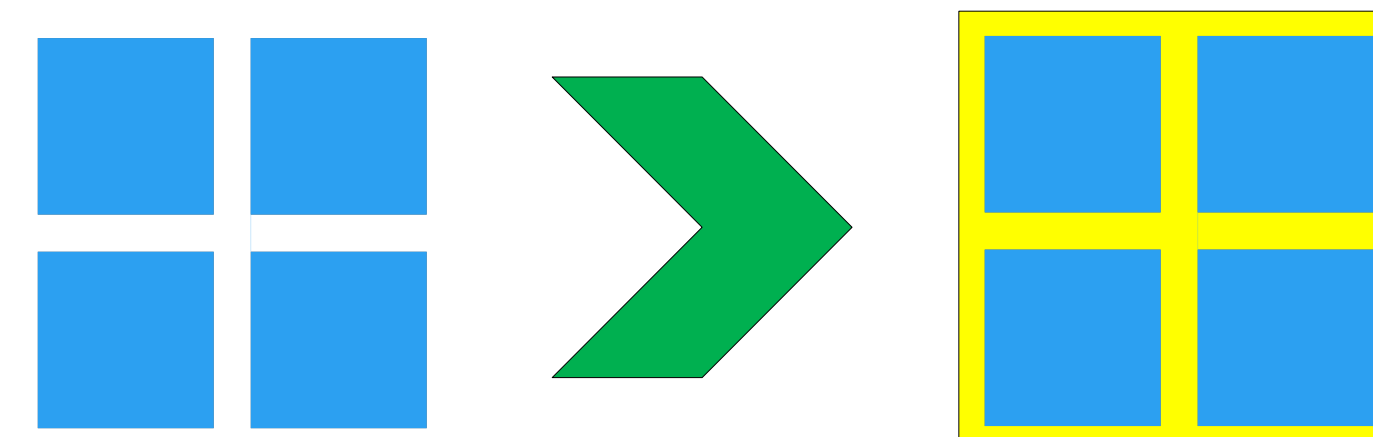


Virtualization Concepts

The creation of many virtual resources from one physical resource



The creation of one virtual resource from one or more physical resource.



Virtualization Types

Application virtualization

This is a process where applications are virtualized and delivered from a server to the end user's device, such as laptops, smartphones, and tablets.

Hardware virtualization

Perhaps the most common type of virtualization today, hardware virtualization is made possible by a virtual machine manager (VM) called the “hypervisor”.

Storage virtualization

This type of virtualization is very easy and cost-effective to implement, since it involves compiling your physical hard drives into a single cluster.

Desktop virtualization

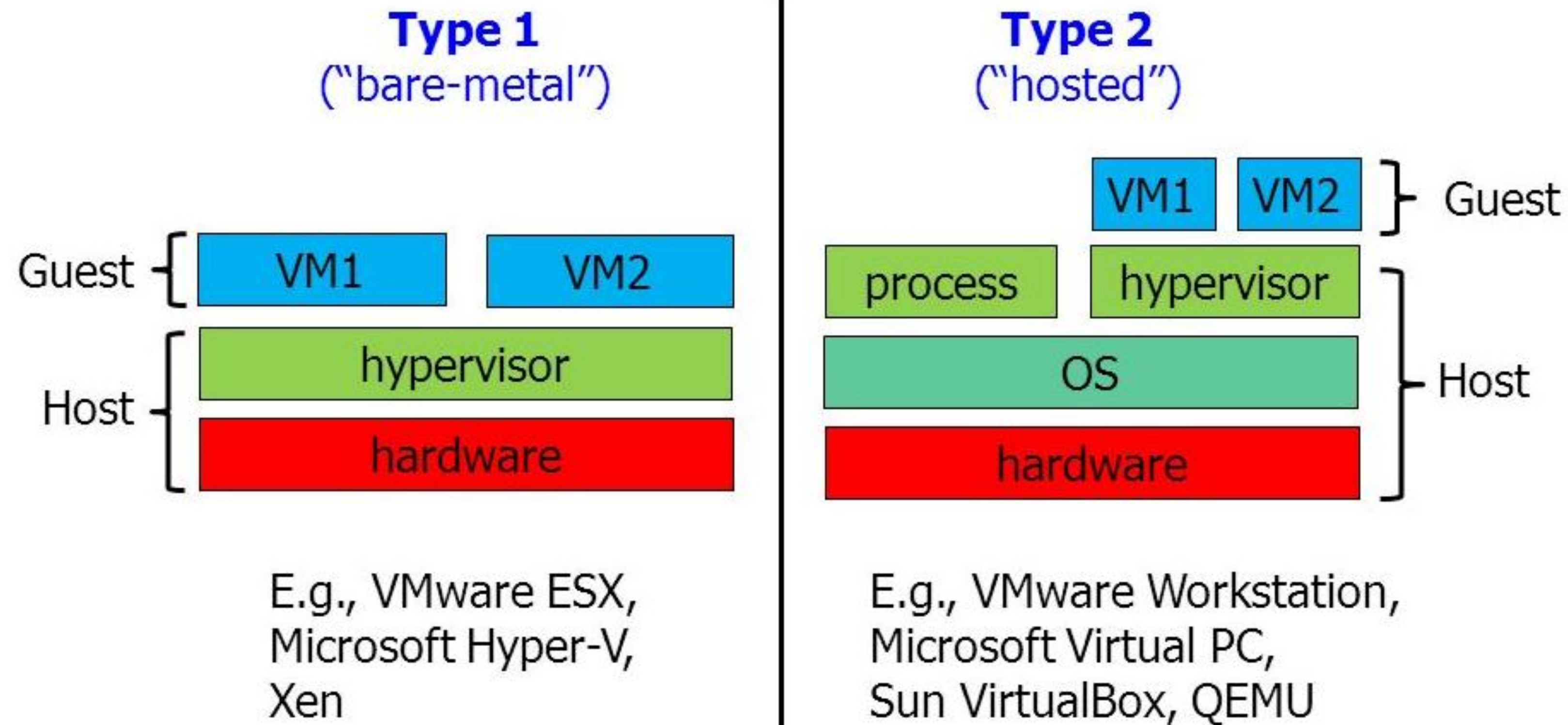
Similar to application virtualization mentioned above, desktop virtualization separates the desktop environment from the physical device and configured as a “virtual desktop infrastructure” (VDI).

Network virtualization

Network virtualization combines all physical networking equipment into a single, software-based resource. It also divides available bandwidth into multiple, independent channels, each of which can be assigned to servers and devices in real time.



Hypervisors



Type1

VMware ESXi

Is a hypervisor designed for hardware virtualization. ESXi installs directly onto a server and has direct control over a machine's underlying resources. ESXi will run without an OS and includes its own kernel. ESXi is the compact, and now preferred, version of VMware's ESX. ESXi is smaller and doesn't contain the ESX service console.

Microsoft Hyper-V

Is a hypervisor designed for hardware virtualization on an x86 architecture. Hyper-V isolates VMs in partitions, where each guest OS will execute a partition. Partitions operate in the manner of parent and child partitions. Parent partitions have direct access to the hardware, while child partitions have a virtual view of system resources. Parent partitions create child partitions using a hypercall API. Hyper-V is available for 64-bit versions of Windows 8 Professional, Enterprise, Education and later.

Xen

Is an open source hypervisor. Xen is included in the Linux kernel and is managed by the Linux Foundation. However, Xen is only supported by a small amount of Linux distributions, such as SUSE Linux Enterprise Server. The software supports full virtualization, paravirtualization and hardware-assisted virtualization. XenServer is another open source Xen product to deploy, host and manage VMs.



Type2

VMware Workstation/Fusion/Player

VMware Player is a free virtualization hypervisor.

It is intended to run only one virtual machine (VM) and does not allow creating VMs.

VMware Server

VMware Server is a free, hosted virtualization hypervisor that's very similar to the VMware Workstation.

KVM

This is a virtualization infrastructure for the Linux kernel. It supports native virtualization on processors with hardware virtualization extensions.

The open-source KVM (or Kernel-Based Virtual Machine) is a Linux-based type-1 hypervisor that can be added to most Linux operating systems including Ubuntu, Debian, SUSE, and Red Hat Enterprise Linux, but also Solaris, and Windows.

Microsoft Virtual PC

This is the latest Microsoft's version of this hypervisor technology, Windows Virtual PC and runs only on Windows 7 and supports only Windows operating systems running on it.

Oracle VM VirtualBox

VirtualBox hypervisor technology provides reasonable performance and features if you want to virtualize on a budget. Despite being a free, hosted product with a very small footprint, VirtualBox shares many features with VMware vSphere and Microsoft Hyper-V.

Red Hat Enterprise Virtualization

Red Hat's Kernel-based Virtual Machine (KVM) has qualities of both a hosted and a bare-metal virtualization hypervisor. It can turn the Linux kernel itself into a hypervisor so the VMs have direct access to the physical hardware.

Virtual Servers vs. Physical Servers

Pros of Physical Servers

- Your IT team has full access to your dedicated server resources 24/7/365, which can be advantageous for high-demand or business-critical operations.
- Depending on your needs, the server may be located onsite or in a conveniently-located colocation facility.
- Your IT staff can fully customize and configure the server to your specifications and business requirements.

Cons of Physical Servers

- Relatively high cost of server purchase, maintenance, and in the case of hardware failure, replacement.
- Your IT staff will be responsible for setup, configuration, and maintenance on a continual basis.
- You may need to hire a contractor or vendor for help if you have difficulty with configuration or repairs.
- Users cannot scale storage in small increments once you reach the maximum workload, since you purchase and own the hardware.
- Requires physical or colocation space.

Virtual Servers vs. Physical Servers

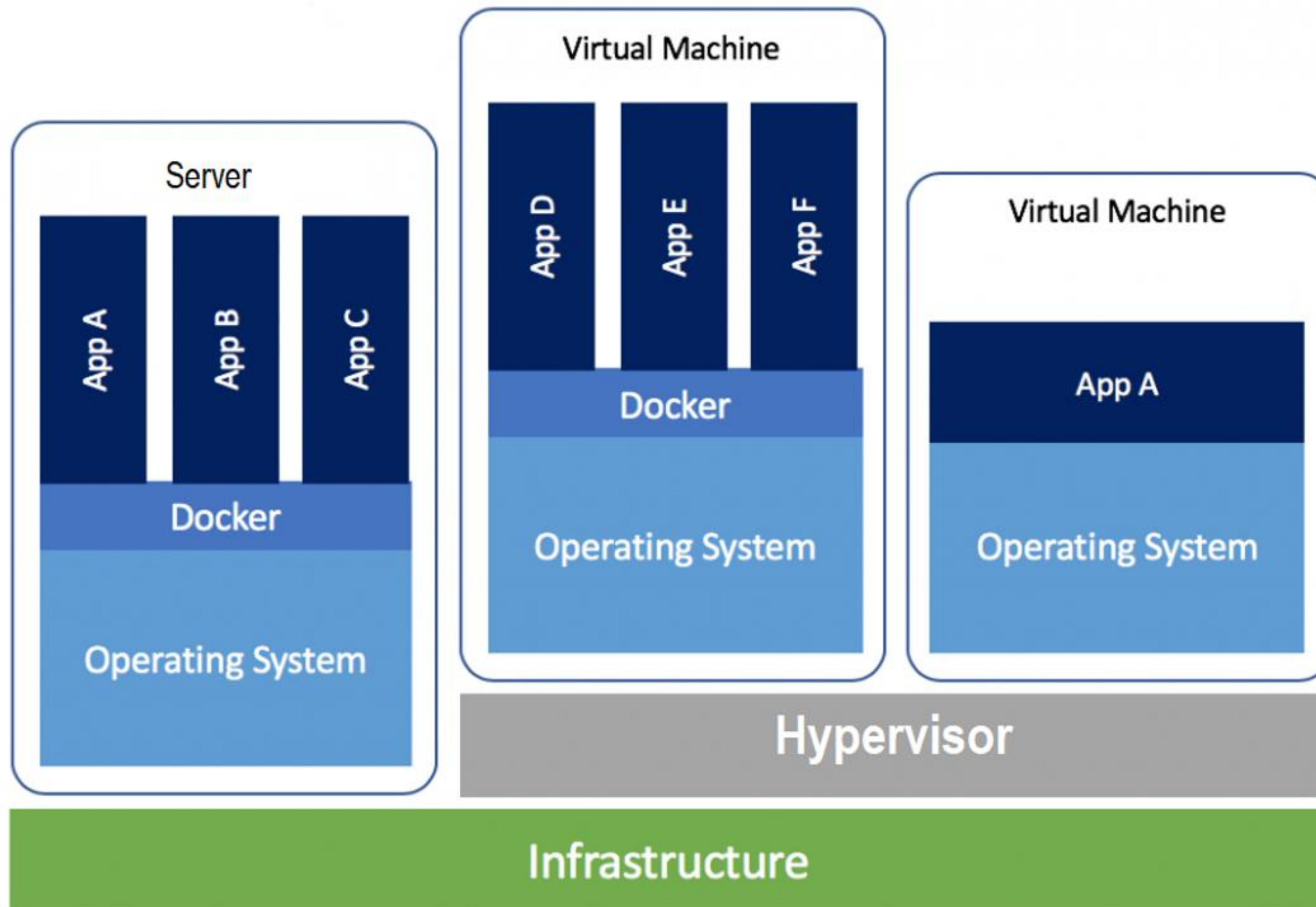
Pros of Virtual Servers

- Fewer upfront costs, since you are not required to purchase hardware.
- Potential for lower lifetime costs on configuration and maintenance.
- Access to expert assistance on setup, configuration, maintenance, and software licenses.
- Reduced need for in-house IT talent.
- Potential to consolidate servers and increase workload efficiency.
- Reduced environmental footprint, due to shared workloads.

Cons of Virtual Servers

- Potential for higher recurring monthly costs, as opposed to higher upfront investment.
- Possible issues with application compatibility, which are best discussed with a knowledgeable vendor.
- Not all vendors are amenable to scaling your agreement up or down in small increments, which largely depends on vendor policies and service level agreements (SLAs).
- You are no longer independently in control of your physical server and the applications running on them.

Are Containers Replacing Virtual Machines?





GET YOUR FANCY KNIFES OUT, WE WILL BE DISCUSSING FINE-DINING HERE ...

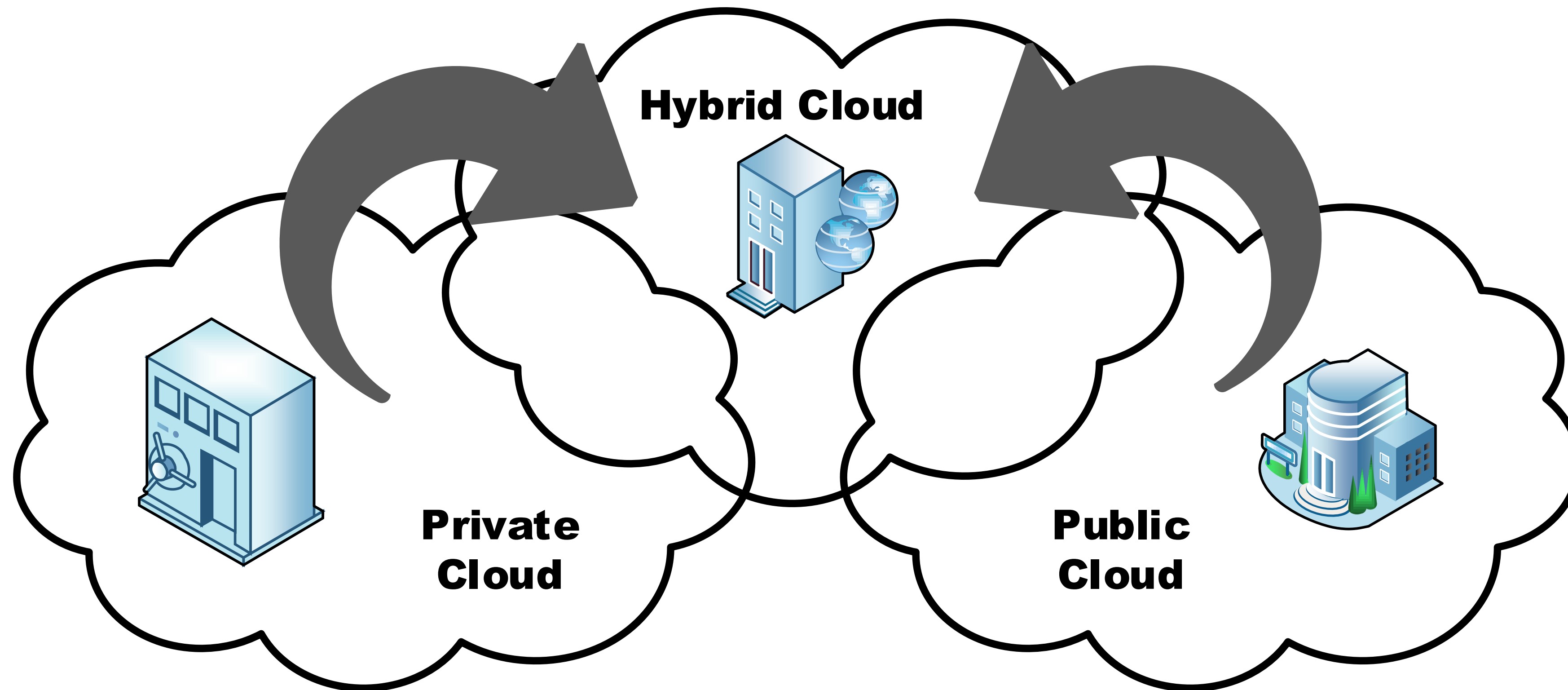
Cloud properties

Why cloud?

- ***On-demand self-service***: Users of clouds can “unilaterally provision computing capabilities, such as server time and network storage, as needed automatically without requiring human interaction with each service provider”.
- ***Availability of “infinite” computing resources***: Cloud users do not have to plan the provision of computing resources in advance, they have the potential to access computing resources by simply making a demand.
- ***Rapid elasticity and adaptability***: Computing resources can be provisioned in an elastic and rapid way by allowing users to buy computing services at any time at various granularities and be able to up and downscale those services according to their needs.
- ***Elimination of up-front commitment***: Users of cloud services do not have to make any heavy, upfront IT investments. This allows companies to start small and to successively increase hardware and software resources only when needed.
- ***Short-term pay for use***: Users are able to pay for their use of cloud services on a short-time basis.
- ***Network access***: Computer services are accessed over the network and through standard mechanisms thus allowing users to connect several devices to the cloud (e.g. laptops, mobile phones).
- ***Pooling of resources***: Providers’ cloud services are pooled to serve multiple customers using a multitenant model, with different physical and virtual resources dynamically assigned and reassigned according to consumer demand.
- ***Measured service and adaptability***: Cloud computing systems are able to react on-time to changes in the amount of computing resources requested and thus automatically control and leverage resources.



Cloud deployment models



Cloud vendors



PaaS vs IaaS vs SaaS



Infrastructure as a service (IaaS)

Infrastructure as a Service is the most flexible category of cloud services. It aims to give you complete control over the hardware that runs your application (IT infrastructure servers and virtual machines (VMs), storage, networks, and operating systems). Instead of buying hardware, with IaaS, you rent it. It's an instant computing infrastructure, provisioned and managed over the internet.



Platform as a service (PaaS)

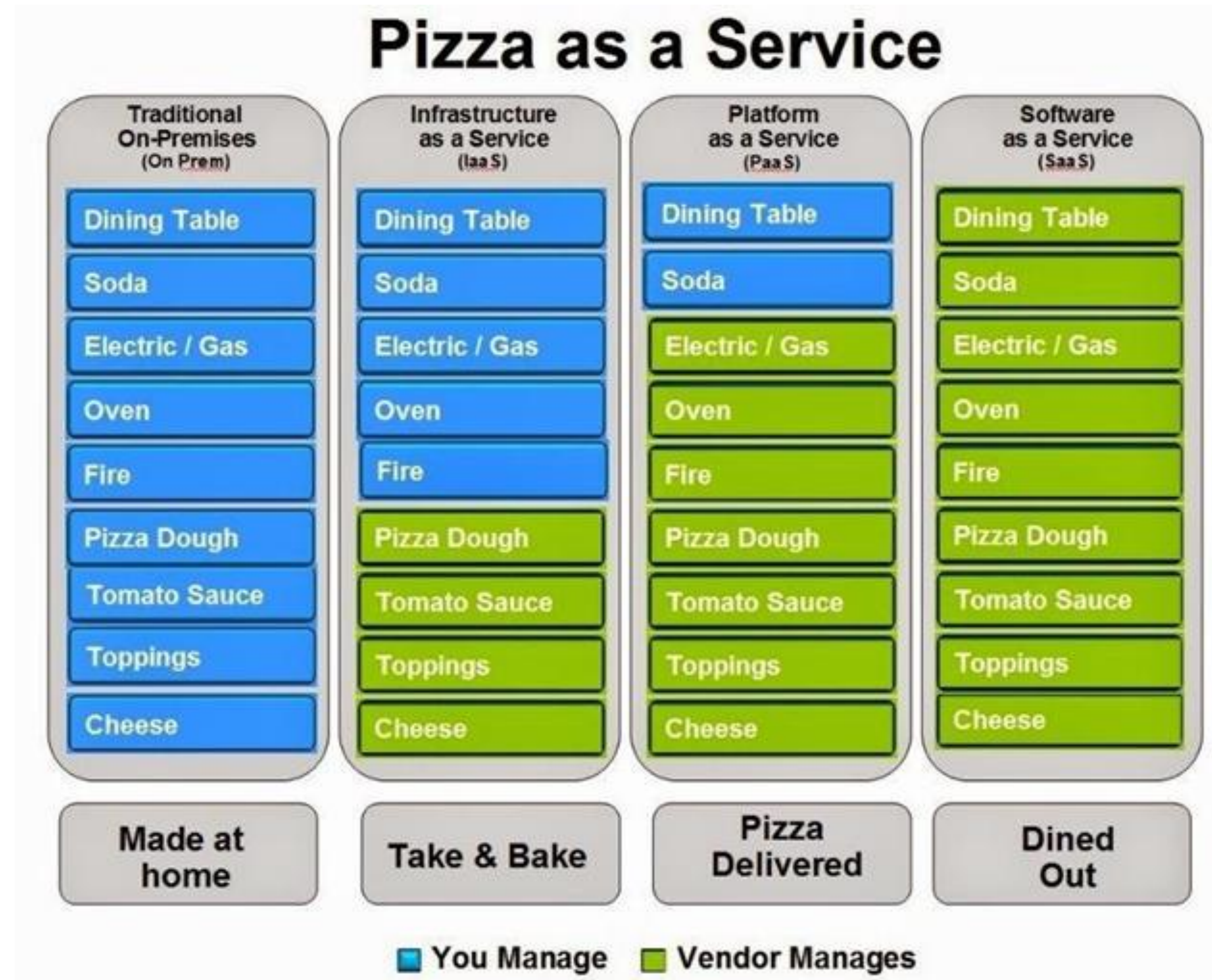
PaaS provides an environment for building, testing, and deploying software applications. The goal of PaaS is to help you create an application quickly without managing the underlying infrastructure. For example, when deploying a web application using PaaS, you don't have to install an operating system, web server, or even system updates.



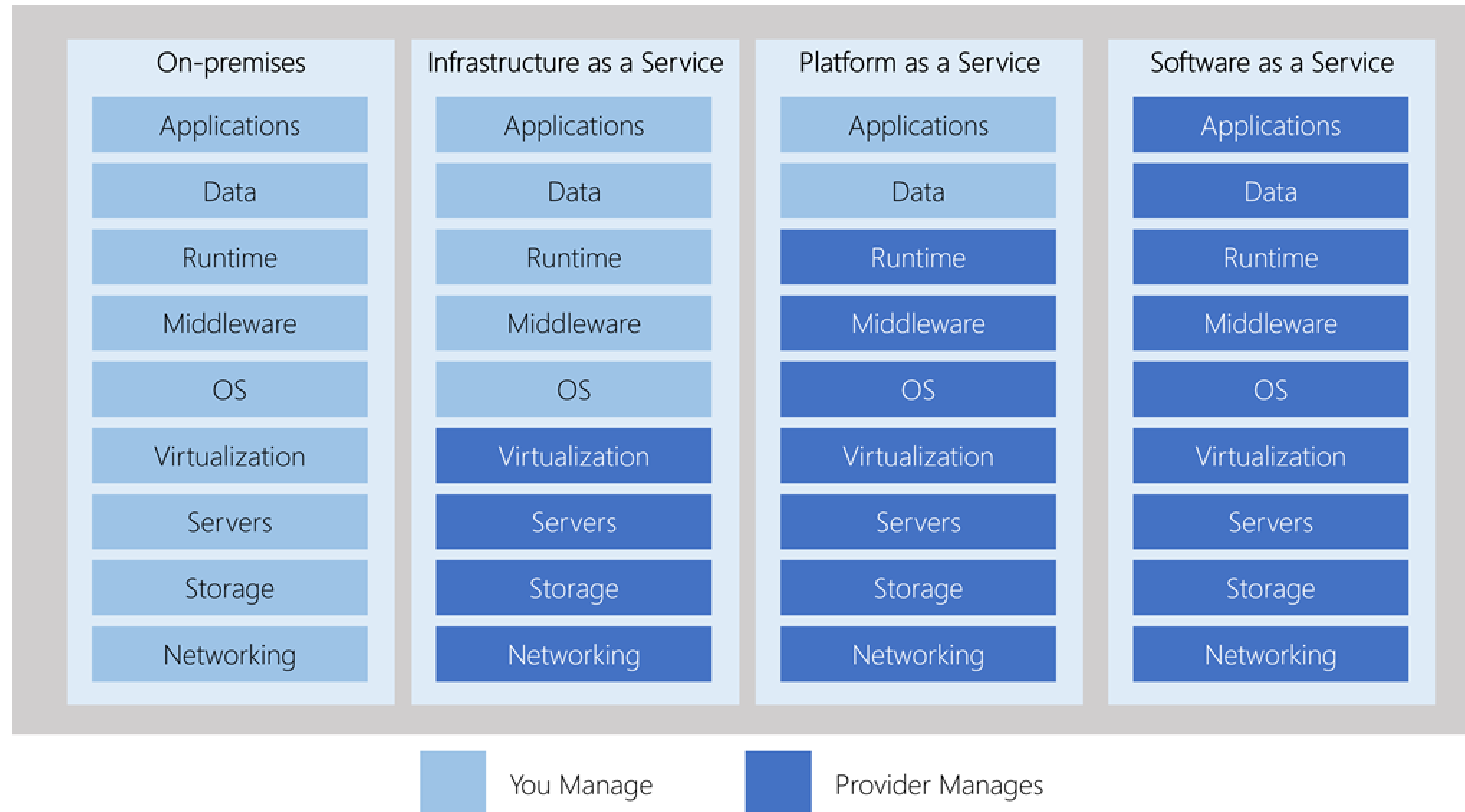
Software as a service (SaaS)

SaaS is software that is centrally hosted and managed for the end customer. It is usually based on an architecture where one version of the application is used for all customers, and licensed through a monthly or annual subscription. Office 365, Skype, and Dynamics CRM Online are perfect examples of SaaS software.

PaaS vs IaaS vs SaaS



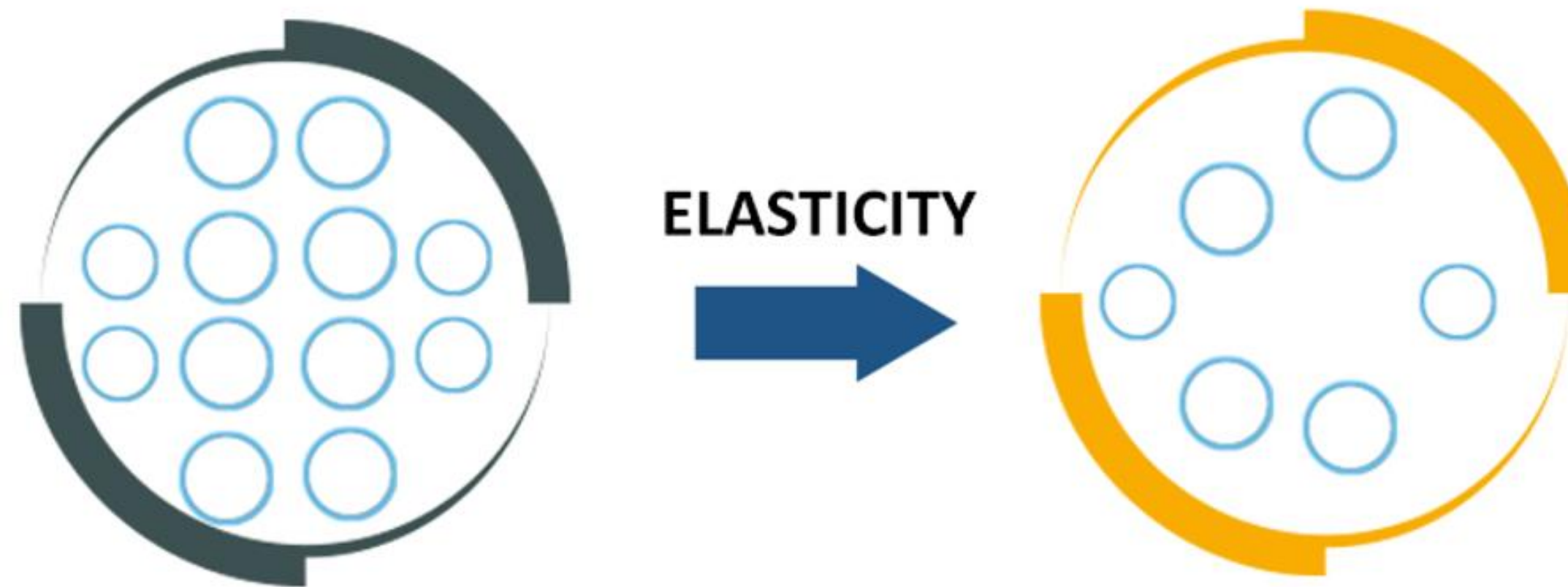
PaaS vs IaaS vs SaaS



Cloud Elasticity

Why?

- Constantly fluctuating loads
- Cyclical loads
- Unpredictable bursts



✓ *Elasticity is enacted by scaling cloud resources*

Cloud Scalability

Vertical Scaling (Scaling Up And Down)

- Attempt to increase or decrease the capacity of a resource (e.g. adding CPUs or RAM)

Horizontal Scaling (Scaling In And Out)

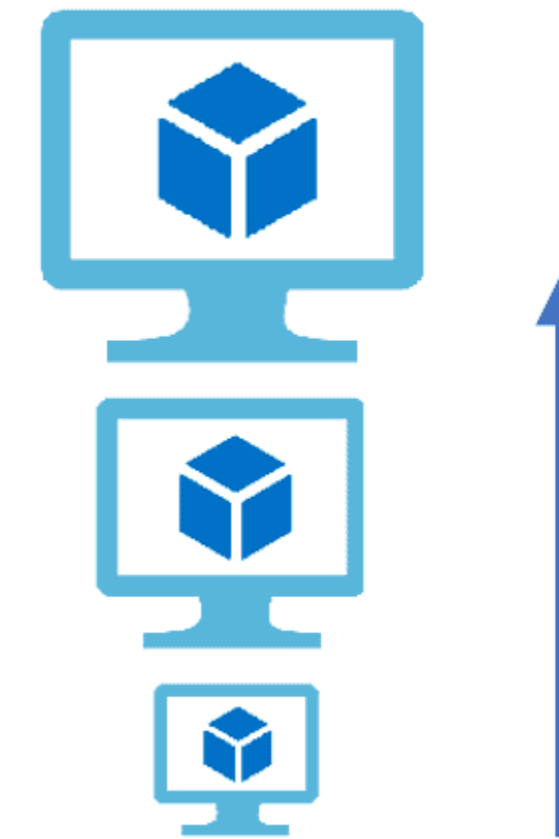
- Strategy whereby additional resources are added to the system or extraneous resources are removed from the system over time

Autoscaling Mechanisms

- Time-based
- Metrics-based

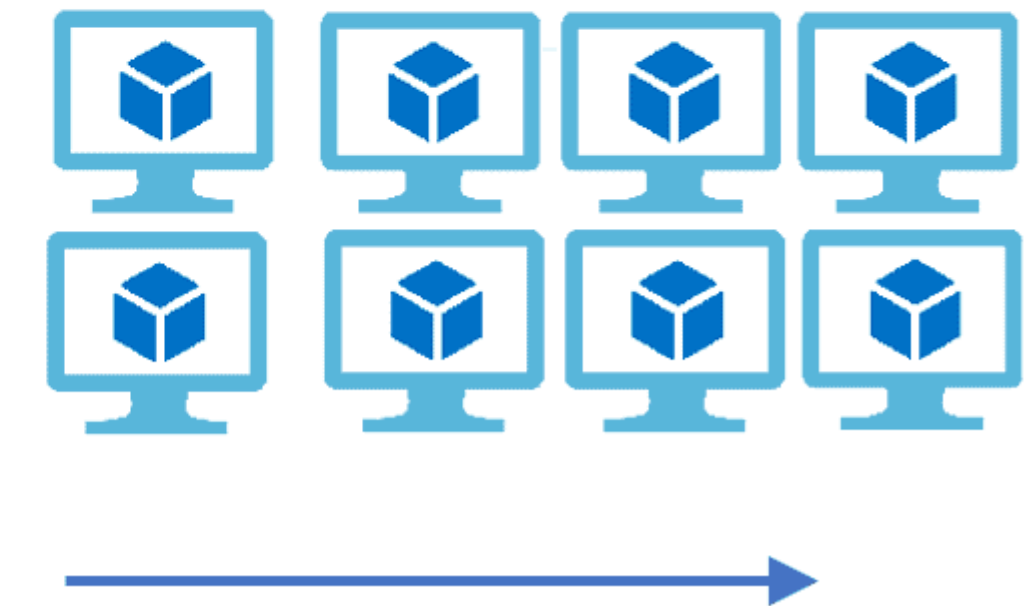
Vertical Scaling

(Increase size of instance (RAM , CPU etc.))



Horizontal Scaling

(Add more instances)

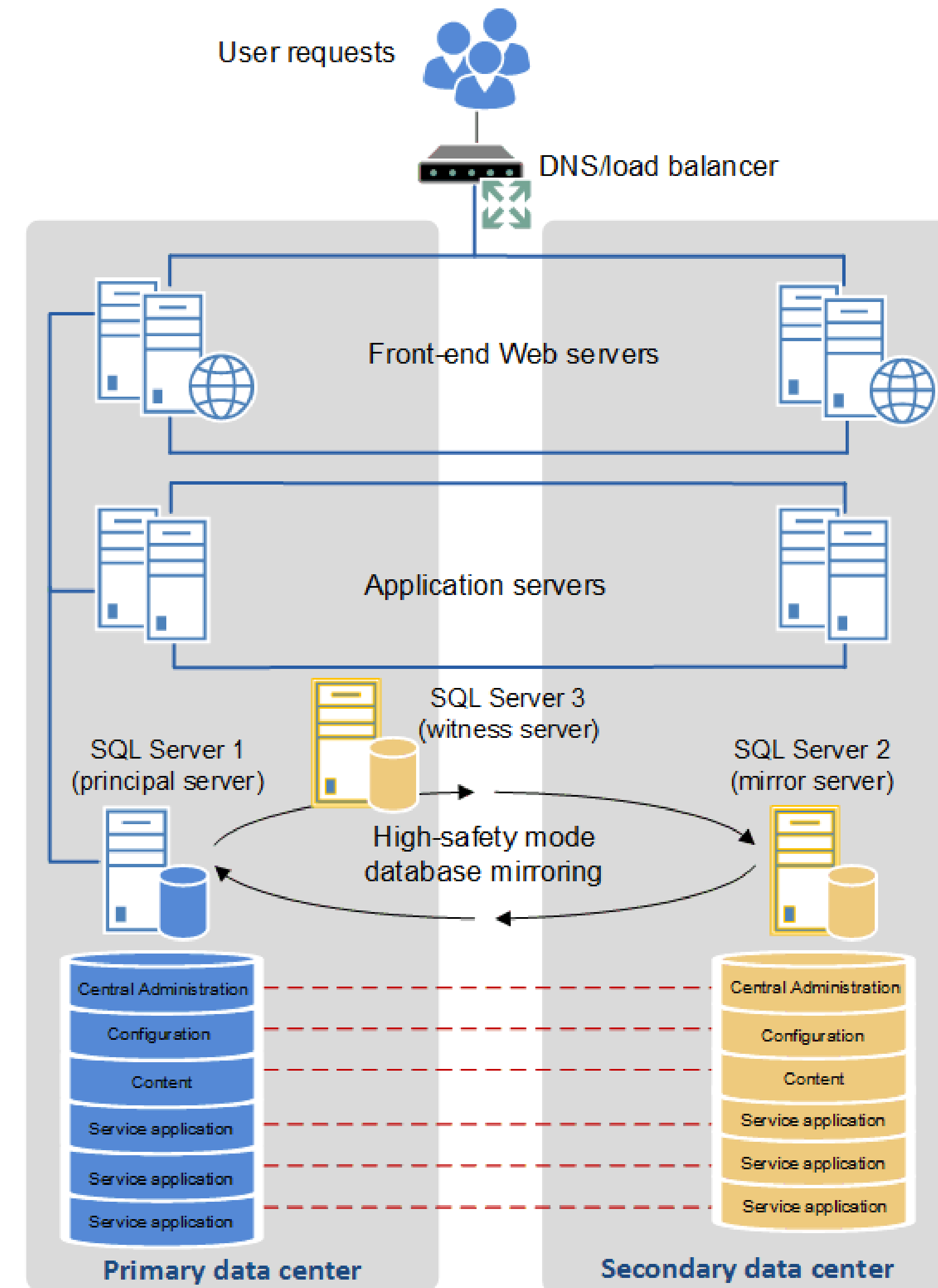


High Availability

High availability is a quality of computing infrastructure that allows it to continue functioning, even when some of its components fail.

The basic elements of high availability:

- **Redundancy** - ensuring that any elements critical to system operations have an additional, redundant component that can take over in case of failure.
- **Monitoring** - collecting data from a running system and detecting when a component fails or stops responding.
- **Failover** - a mechanism that can switch automatically from the currently active component to a redundant component, if monitoring shows a failure of the active component



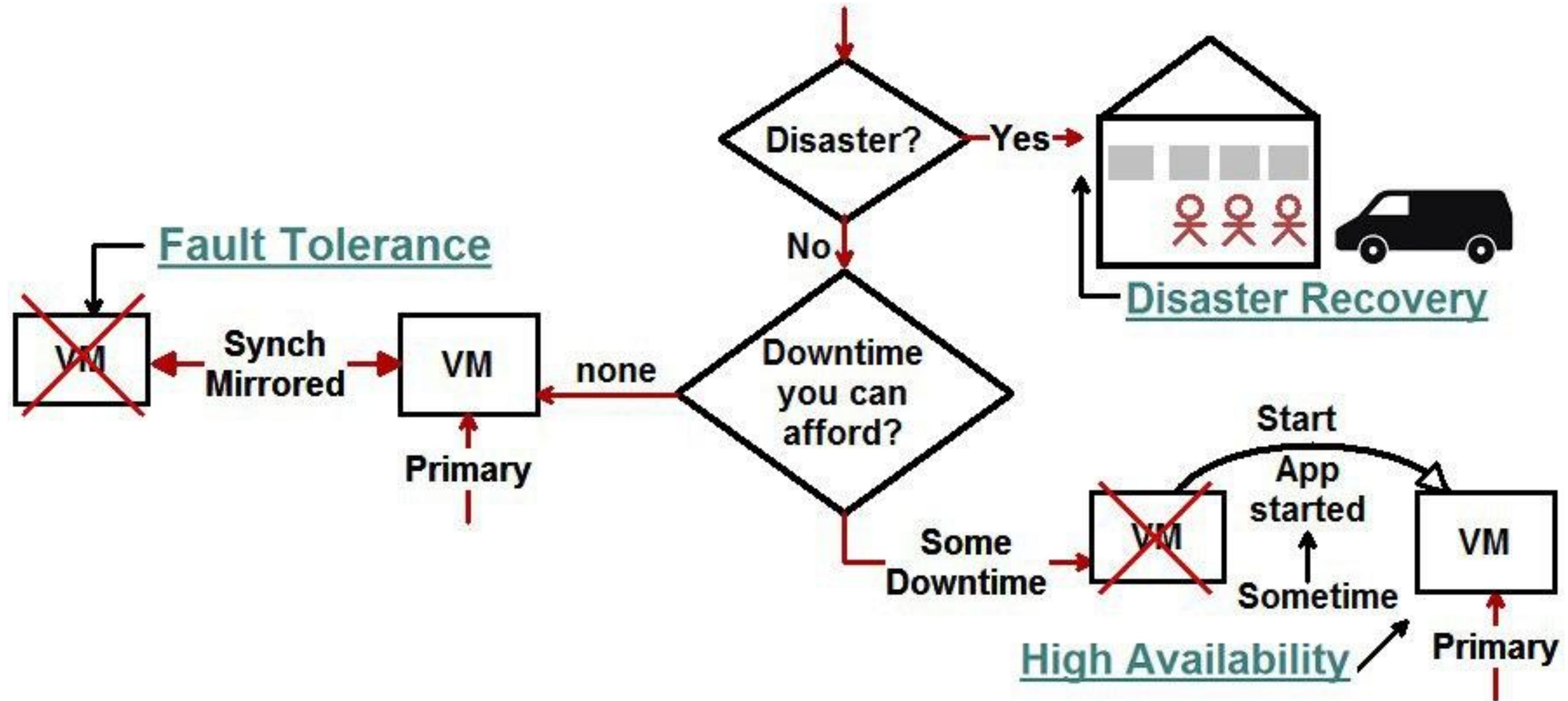
Fault Tolerance And Disaster Recovery

*FT - extremely similar to HA but goes one step further by **guaranteeing zero downtime**.*

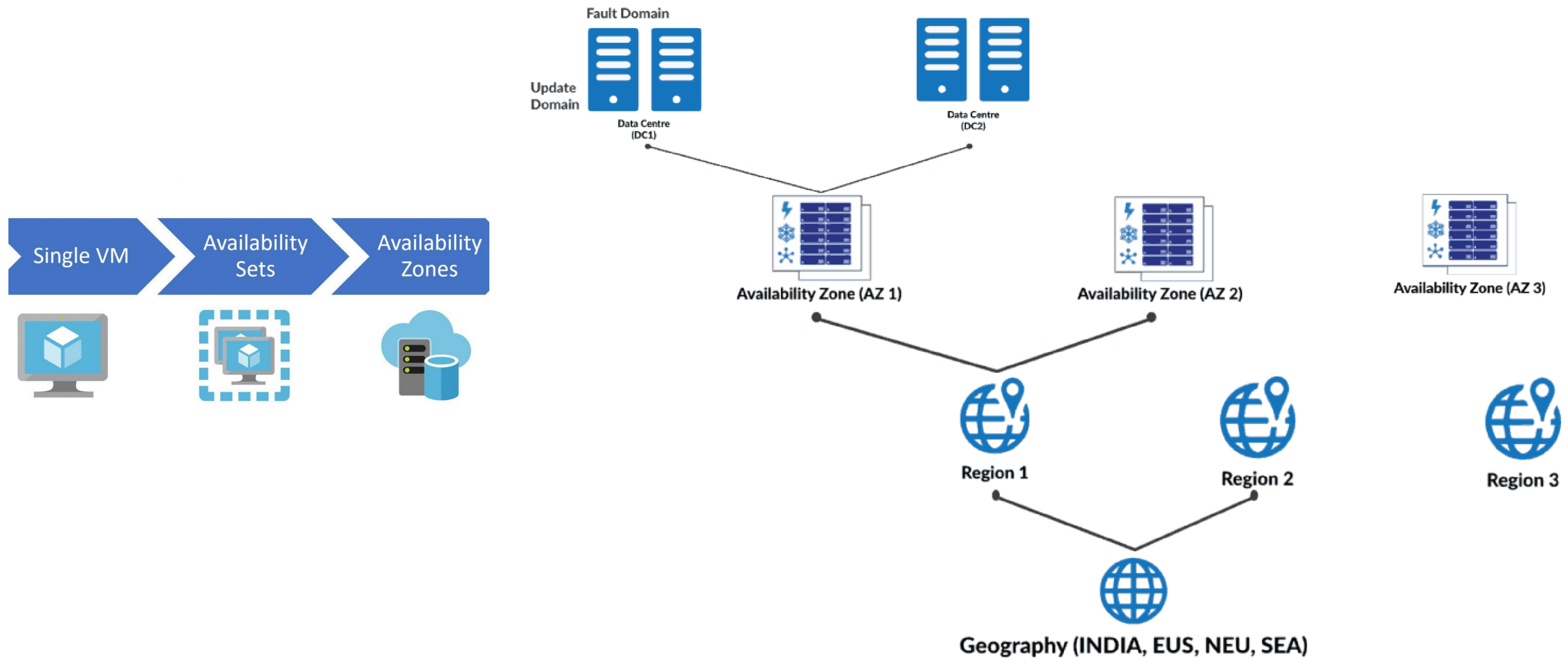
*DR goes beyond FT or HA and consists of a **complete plan to recover critical business systems and normal operations** in the event of a catastrophic disaster.*



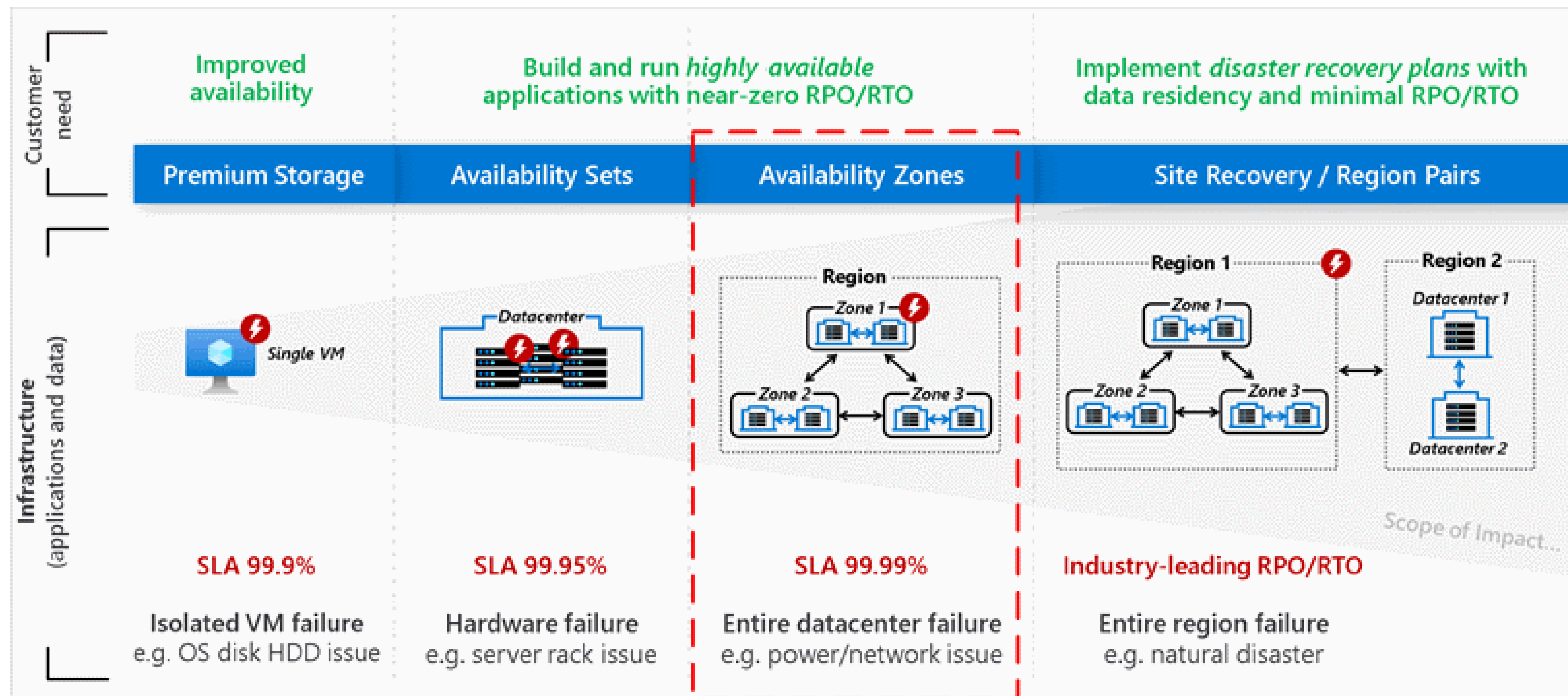
HA vs FT vs DR



Regions and availability zones



Regions and availability zones





FOUND THESE LYING AROUND THE HOUSE SO I THOUGHT YOU MIGHT FIND THEM USEFUL

Resources



- <https://www.onlinecomputertips.com/support-categories/hardware/463-server-virtualization-explained>
- <https://www.youtube.com/watch?v=NmrJIWDG0yk>
- <https://www.youtube.com/watch?v=RsSiewDb7Dk>
- <https://www.youtube.com/watch?v=1WnDHitznGY>
- <https://www.youtube.com/watch?v=uroryFU78gM>
- <https://www.youtube.com/watch?v=usYySG1nbfl>
- <https://cloudacademy.com/course/what-is-cloud-computing-introductory/key-cloud-concepts/>
- Homework ->



For this lesson you will get the chance to spin up your own “local cloud” on your PC. Using virtualization tools, you will get hands on experience on some of the core features of the cloud: Reliability, Scalability, Elasticity and Fault tolerance.

Prerequisite for this homework is having VirtualBox installed. Please use the following link to download VirtualBox on your PC:

<https://endava.sharepoint.com/Disciplines/itsms/ITSM%20Tools/Forms/DispForm.aspx?ID=465>.

Tasks

- Create 3 Linux virtual machines in VirtualBox. The choice for OS distribution is entirely up to you.

Note: Do not make the VMs too large. 1 CPU, 8GB disk space and 512MB RAM should suffice.

- Deploy a static HTML page on port 80 that explains the steps you took to finish the tasks from this homework, using Apache2 or Nginx.

Note: You are free to use previously learned tools for automating this setup (Ansible for configuration management, Docker for containerization, etc.)

- Optionally, add a load balancer to one instance, that will randomly route the traffic to one of the three servers on each request (Add an ID for each VM in the HTML content, to recognize on which server the LB routed).

- Now, scale in to 2 VMs. Document this process in the HTML files (locally, without deploying them to the servers). There should be no service downtime.

Note: If you did the optional step, stop an instance that is not a load balancer.

- Scale up the existing VMs to 2 CPUs and 1GB RAM. Also document this process in the local HTML files.
- Now perform a deployment of the local HTML files to the servers.
- Scale out back to 3 VMs, by starting the instance you previously stopped. Make sure this instance has the latest HTML deployed.
- Scale down the two instances that you previously scaled up, back to 1 CPU and 512MB RAM.
- After documenting everything, deploy the latest version of the HTML page one more time.

If your are really bored... try and fix this:

<https://github.com/Biohazardhpk/packstack-challenge>